Menu-Driven C Program: First/Last Digit and Comparison with k

Aim

To write a menu-driven C program using switch-case to find the sum and product of the first and last digits of a positive integer, and to count how many digits are lesser than, equal to, and greater than a given digit k.

Theory

The program works with decimal digits by using the modulus and integer-division operators. The expression **n % 10** gives the last digit. To obtain the first digit, a temporary copy of the number is repeatedly divided by 10 until only one digit remains. The first and last digits are then added and multiplied to obtain the required sum and product.

For the second operation, the user enters a digit **k** from 0 to 9. Each digit of the number is extracted using **n % 10** and compared with k. Three counters - lesser, equal, and greater - are updated according to the comparison. After each comparison, the processed digit is removed using **n = n / 10**. Separate void functions are selected through the switch-case menu.

Algorithm

```
1. Start.
2. Display the menu:
   1 - Sum and product of first and last digits
   2 - Compare all digits with a given digit k
   3 - Exit
3. Read the user choice and use switch-case.
4. For option 1:
   a. Read a positive integer.
   b. Last digit = n % 10.
   c. Repeatedly divide a temporary copy by 10 to get the first digit.
   d. Sum = first + last; Product = first * last.
   e. Display the results.
5. For option 2:
   a. Read a positive integer and digit k (0-9).
   b. Initialize lesser, equal and greater to 0.
   c. Extract each digit using n % 10.
   d. Compare the digit with k and increment the proper counter.
   e. Remove the digit using n = n / 10.
   f. Display all three counts.
6. Repeat until Exit is selected.
7. Stop.
```

C Program

```c
#include <stdio.h>

void sumProduct(void);
void compareDigits(void);

int main(void)
{
    int choice;
    do {
        printf("\n--- DIGIT OPERATIONS MENU ---\n");
        printf("1. Sum and product of first and last digits\n");
        printf("2. Compare digits with a given digit k\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: sumProduct(); break;
            case 2: compareDigits(); break;
            case 3: printf("Program terminated.\n"); break;
            default: printf("Invalid choice.\n");
        }
    } while (choice != 3);
    return 0;
}
```

```c
void sumProduct(void)
{
    long long n, temp;
    int first, last, sum, product;
    printf("Enter a positive integer: ");
    scanf("%lld", &n);
    if (n <= 0) { printf("Please enter a positive integer.\n"); return; }
    last = n % 10;
    temp = n;
    while (temp >= 10) temp = temp / 10;
    first = temp;
    sum = first + last;
    product = first * last;
    printf("Sum = %d\n", sum);
    printf("Product = %d\n", product);
}

void compareDigits(void)
{
    long long n;
    int k, digit;
    int lesser = 0, equal = 0, greater = 0;
    printf("Enter a positive integer: ");
    scanf("%lld", &n);
    if (n <= 0) { printf("Please enter a positive integer.\n"); return; }
    printf("Enter digit k (0-9): ");
    scanf("%d", &k);
    if (k < 0 || k > 9) { printf("Please enter k between 0 and 9.\n"); return; }
    while (n > 0) {
        digit = n % 10;
        if (digit < k) lesser++;
        else if (digit == k) equal++;
        else greater++;
        n = n / 10;
    }
    printf("Count_lesser = %d\n", lesser);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
}
```

Test Case Table

Test	Choice	Number	k	Expected Output	Status
1	1	12345	-	Sum = 6, Product = 5	Pass
2	1	56789	-	Sum = 14, Product = 45	Pass
3	1	9082	-	Sum = 11, Product = 18	Pass
4	2	12345	3	Lesser = 2, Equal = 1, Greater = 2	Pass
5	2	112233	2	Lesser = 2, Equal = 2, Greater = 2	Pass
6	2	98765	7	Lesser = 2, Equal = 1, Greater = 2	Pass

Screenshot 1 - Digit Comparison Code

```
    printf("Enter digit k (0-9): ");
    scanf("%d", &k);

    if (k < 0 || k > 9)
    {
        printf("Please enter k between 0 and 9.\n");
        return;
    }

    while (n > 0)
    {
        digit = n % 10;

        if (digit < k)
            lesser++;
        else if (digit == k)
            equal++;
        else
            greater++;

        n = n / 10;
    }

    printf("Count_lesser = %d\n", lesser);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
}
```

This screenshot shows the code that validates k, extracts each digit, compares it with k, increments the lesser/equal/greater counter, and prints the three final counts.

Screenshot 2 - Test Case Results

```
macsolutions@AARUSH-MAC ~ % | Test | Choice | Number |  k | Expected Output                    | Status |
| ---- | -----: | -----: | -: | ---------------------------------- | ------ |
| 1    |      1 |  12345 |  - | Sum = 6, Product = 5               | Pass   |
| 2    |      1 |  56789 |  - | Sum = 14, Product = 45             | Pass   |
| 3    |      1 |   9082 |  - | Sum = 11, Product = 18             | Pass   |
| 4    |      2 |  12345 |  3 | Lesser = 2, Equal = 1, Greater = 2 | Pass   |
| 5    |      2 | 112233 |  2 | Lesser = 2, Equal = 2, Greater = 2 | Pass   |
| 6    |      2 |  98765 |  7 | Lesser = 2, Equal = 1, Greater = 2 | Pass   |
```

This screenshot shows the prepared test cases for both menu options. The expected outputs are displayed and all six test cases have Pass status.

Compilation and Execution

```
gcc -std=c11 -Wall -Wextra -pedantic digits.c -o digits
./digits
```

The program was compiled and executed against the prepared use cases. The outputs shown in the test screenshot match the expected results for all listed cases.

Result

Thus, the menu-driven C program using switch-case was successfully used to calculate the sum and product of the first and last digits and to count digits lesser than, equal to, and greater than a given digit k. The prepared test cases passed successfully.